

OptiCrop: Smart Agricultural Production Optimization Engine

Project Description:

The Crop Recommendation System is a Machine Learning-based web application that helps farmers choose the most suitable crop based on soil nutrients and environmental conditions. The application predicts the best crop by analyzing important agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, soil pH, and rainfall.

The system is developed using Python, Flask, HTML, CSS, and Machine Learning. A Logistic Regression model is trained on agricultural datasets and saved using Pickle. The Flask backend receives user inputs, processes them through the trained model, and returns the recommended crop instantly.

The application provides an easy-to-use interface where users can enter field conditions and receive accurate crop recommendations. This helps improve agricultural productivity, reduce crop failure, and support data-driven farming decisions.

Scenarios

Scenario 1:

A farmer enters the following soil details:

- Nitrogen: 90
- Phosphorus: 42
- Potassium: 43
- Temperature: 21°C
- Humidity: 82%
- pH: 6.5
- Rainfall: 202 mm

The system predicts **Rice** as the most suitable crop based on the given environmental conditions.

Scenario 2:

A farmer from a dry region enters:

- Nitrogen: 70
- Phosphorus: 50
- Potassium: 40
- Temperature: 30°C
- Humidity: 55%
- pH: 7.1
- Rainfall: 75 mm

The application recommends **Maize**, which grows well under these climatic conditions.

Scenario 3:

A farmer provides:

- Nitrogen: 35
- Phosphorus: 25
- Potassium: 30
- Temperature: 27°C
- Humidity: 65%
- pH: 6.8
- Rainfall: 110 mm

The system predicts **Cotton** as the best crop and displays the recommendation within seconds.

Technical Architecture

Description

The Crop Recommendation System follows a **three-layer architecture**.

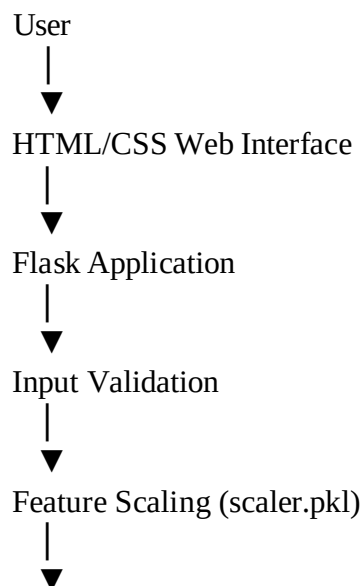
The **Presentation Layer** consists of HTML, CSS, and JavaScript pages that collect user inputs through an interactive web interface.

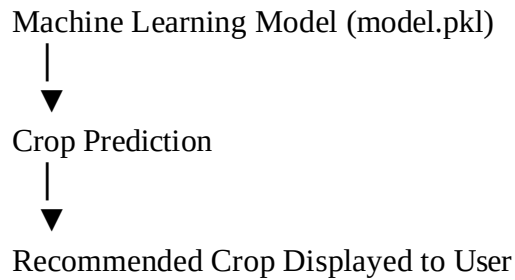
The **Application Layer** is developed using Flask. It validates the user inputs, loads the trained machine learning model (model.pkl) and scaler (scaler.pkl), processes the data, and performs prediction.

The **Machine Learning Layer** contains the trained Logistic Regression model built using Scikit-learn. The model predicts the most suitable crop based on the input parameters.

The prediction result is sent back to the Flask application and displayed on the webpage.

Architecture Flow





Pre-requisites

Before developing and running the Crop Recommendation System, ensure that the following software, tools, libraries, and technologies are installed and understood:

- **Python 3.10 or later** – Used for implementing the machine learning model and backend logic.
- **Flask Framework** – Used to develop the web application and handle client-server communication.
- **Machine Learning Concepts** – Basic understanding of classification algorithms, model training, and prediction.
- **Scikit-learn (sklearn)** – Used for data preprocessing, model training, evaluation, and prediction.
- **Pandas** – Used for loading, cleaning, and manipulating the agricultural dataset.
- **NumPy** – Used for numerical computations and array operations.
- **Matplotlib & Seaborn** – Used for Exploratory Data Analysis (EDA) and data visualization.
- **Jupyter Notebook** – Used for data preprocessing, model building, and performance evaluation.
- **HTML5** – Used to create the web pages and user interface.
- **CSS3** – Used for styling and designing the web application.
- **JavaScript** – Used to improve user interaction and form validation.
- **Visual Studio Code (VS Code)** – Used as the development environment for coding and debugging.
- **Git & GitHub** – Used for version control and project management.
- **Pickle** – Used to save and load the trained machine learning model (model.pkl) and scaler (scaler.pkl).
- **Web Browser (Google Chrome/Microsoft Edge)** – Used to access and test the Flask web application.
- **Windows/Linux Operating System** – Supported platform for development and deployment.

Required Python Libraries

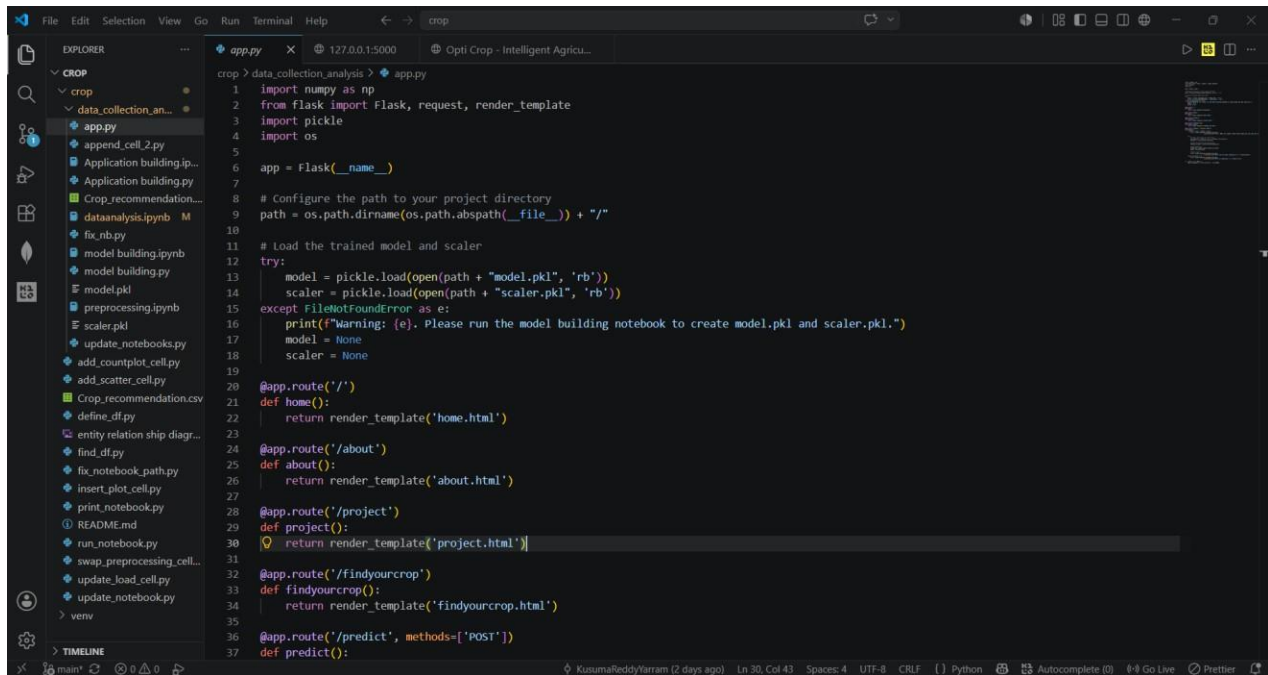
pip install flask
pip install pandas
pip install numpy
pip install scikit-learn
pip install matplotlib
pip install seaborn
pip install joblib

Milestone 1: Project Setup

Step 1: Project Structure

Content:

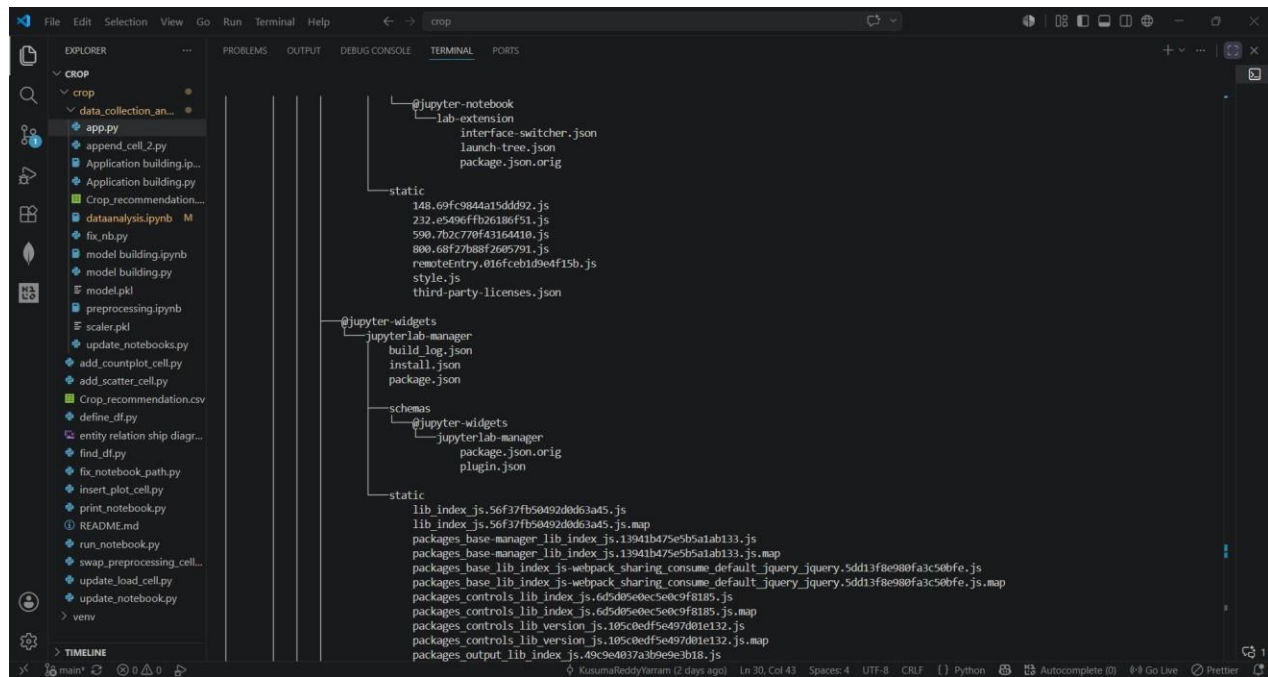
The Crop Recommendation System project was created using Python and Flask. The project contains the machine learning model, dataset, templates, and application files. The folder structure is organized to separate the backend, frontend, and machine learning resources.



The screenshot shows a VS Code editor with a project named 'Opti Crop - Intelligent Agriculture'. The Explorer sidebar on the left displays the project structure, including folders for 'crop', 'data_collection_analysis', and 'app.py'. The main editor area shows the contents of 'app.py', which is a Flask application. The code includes imports for numpy, flask, pickle, and os. It defines a Flask app, configures the path to the project directory, and loads a trained model and scaler. The app has several routes: a home page, an about page, a project page, a findyourcrop page, and a predict page. The predict page uses the loaded model and scaler to make predictions based on user input.

```
1 import numpy as np
2 from flask import Flask, request, render_template
3 import pickle
4 import os
5
6 app = Flask(__name__)
7
8 # Configure the path to your project directory
9 path = os.path.dirname(os.path.abspath(__file__)) + "/"
10
11 # Load the trained model and scaler
12 try:
13     model = pickle.load(open(path + "model.pkl", 'rb'))
14     scaler = pickle.load(open(path + "scaler.pkl", 'rb'))
15 except FileNotFoundError as e:
16     print(f"Warning: {e}. Please run the model building notebook to create model.pkl and scaler.pkl.")
17     model = None
18     scaler = None
19
20 @app.route('/')
21 def home():
22     return render_template('home.html')
23
24 @app.route('/about')
25 def about():
26     return render_template('about.html')
27
28 @app.route('/project')
29 def project():
30     return render_template('project.html')
31
32 @app.route('/findyourcrop')
33 def findyourcrop():
34     return render_template('findyourcrop.html')
35
36 @app.route('/predict', methods=['POST'])
37 def predict():
```

Step 2: Create Virtual Environment



Step 3: Install Required Libraries

Command:

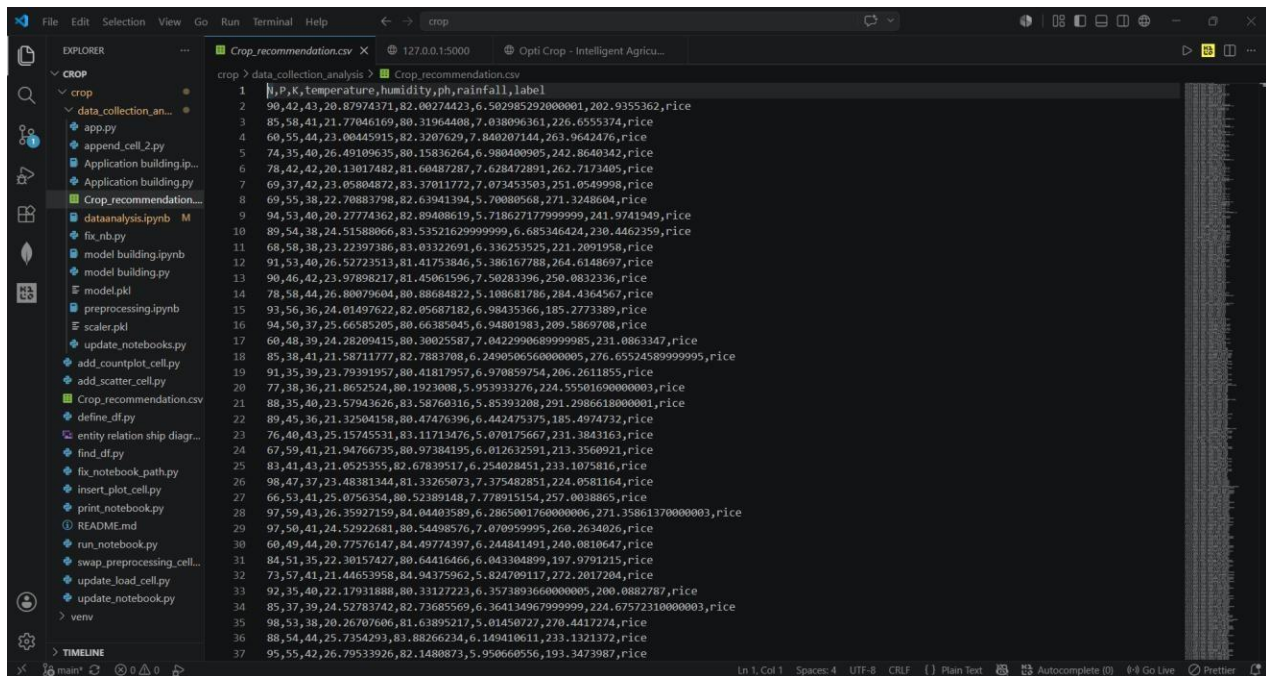
pip install flask pandas numpy scikit-learn matplotlib seaborn

Milestone 2: Machine Learning Model

Step 1: Dataset

Content:

The crop dataset contains soil nutrients and environmental parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, Rainfall, and Crop Label. This dataset is used to train the machine learning model.



```

1  h,p,k,temperature,humidity,ph,rainfall,label
2  90,42,43,20.87974771,82.80274423,6.502985292000001,202.9355362,rice
3  85,50,41,21.77046169,80.31964408,7.038096361,226.6555374,rice
4  60,55,40,23.00445015,82.3207629,7.840207144,263.9642476,rice
5  74,35,40,26.49109635,80.15836264,6.980400905,242.8640342,rice
6  78,42,42,20.13017482,81.60487287,7.628472891,262.7173405,rice
7  69,37,42,23.05804872,83.37011772,7.073453503,251.0549998,rice
8  69,55,38,22.70883798,82.63941394,5.70080568,271.3248604,rice
9  94,53,40,20.27774362,82.89480619,5.718627177999999,241.9741949,rice
10 89,54,38,24.51588066,83.53521629999999,6.685346424,230.4462359,rice
11 68,58,38,23.22397386,83.03322691,6.336253525,221.2091958,rice
12 91,53,40,26.52723513,81.41753846,5.386167788,264.6148697,rice
13 90,46,42,23.97898217,81.45061596,7.50283396,250.0832336,rice
14 78,58,44,26.80079604,80.88684822,5.108681786,284.4364567,rice
15 93,56,36,24.01497622,82.05687182,6.98435366,185.2773389,rice
16 94,50,37,25.66585205,80.66385045,6.94801983,209.5869708,rice
17 60,48,39,24.28209415,80.30025587,7.042299068999999,231.0863347,rice
18 85,38,41,21.58711777,82.7883788,6.2490506560000005,276.65524589999995,rice
19 91,35,39,23.79391957,80.41817957,6.970859754,206.2611855,rice
20 77,38,36,21.8652524,80.1923008,5.953933276,224.55501690000003,rice
21 88,35,40,23.57943626,83.58760316,5.85393208,291.2986618000001,rice
22 89,45,36,21.32504158,80.47476396,6.442475375,185.4974732,rice
23 76,40,43,25.15745531,83.11713476,5.070175667,231.3843163,rice
24 67,59,41,21.94766735,80.97384195,6.012632591,213.3560921,rice
25 83,41,43,21.0525355,82.67839517,6.254028451,233.1075816,rice
26 98,47,37,23.48381344,81.33265073,7.375482851,224.0581164,rice
27 66,53,41,25.0756354,80.52389148,7.778915154,257.0038865,rice
28 97,59,43,26.35927159,84.04403589,6.2865001760000006,271.35861370000003,rice
29 97,50,41,24.52922681,80.54498576,7.070959995,260.2634026,rice
30 60,49,44,20.77576147,84.49774397,6.244841491,240.0810647,rice
31 84,51,35,22.30157427,80.64416466,6.043304899,197.9791215,rice
32 73,57,41,21.44653958,84.94375962,5.824709117,272.2017204,rice
33 92,35,40,22.17931888,80.33127223,6.3573893660000005,200.0882787,rice
34 85,37,39,24.52783742,82.73685569,6.364134967999999,224.67572310000003,rice
35 98,53,38,20.26707606,81.63895217,5.01450727,270.4417274,rice
36 88,54,44,25.7354293,83.88266234,6.149410611,233.1321372,rice
37 95,55,42,26.79533926,82.1488873,5.950660556,193.3473987,rice
  
```

Step 2: Model Building

Content:

The Logistic Regression algorithm is used to train the crop recommendation model. After training, the model is saved as model.pkl and the scaler is saved as scaler.pkl for future predictions.

Milestone 3: Flask Application Development

Step 1: Backend Development

Content:

The Flask application loads the trained machine learning model and provides routes for the Home, About, Project, and Crop Prediction pages.

Code

```
from flask import Flask
```

```
app = Flask(__name__)
```

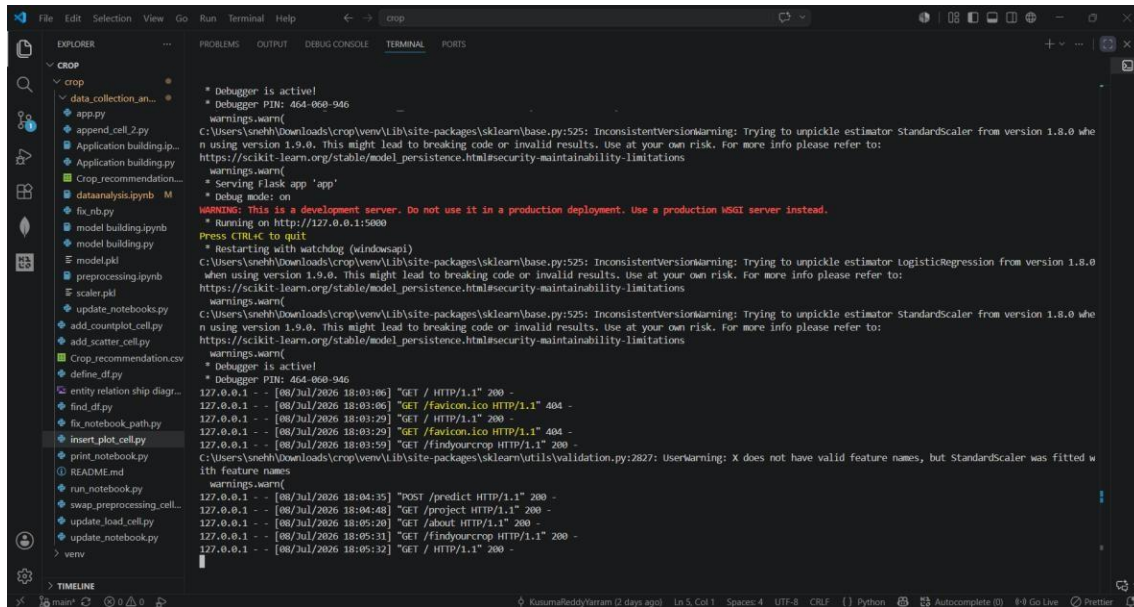
```
@app.route("/")
```

```
def home():
```

```
    return render_template("home.html")
```

```
@app.route("/predict", methods=["POST"])
def predict():
```

Step 2: Run the Application



Milestone 4: Website Interface

Home Page

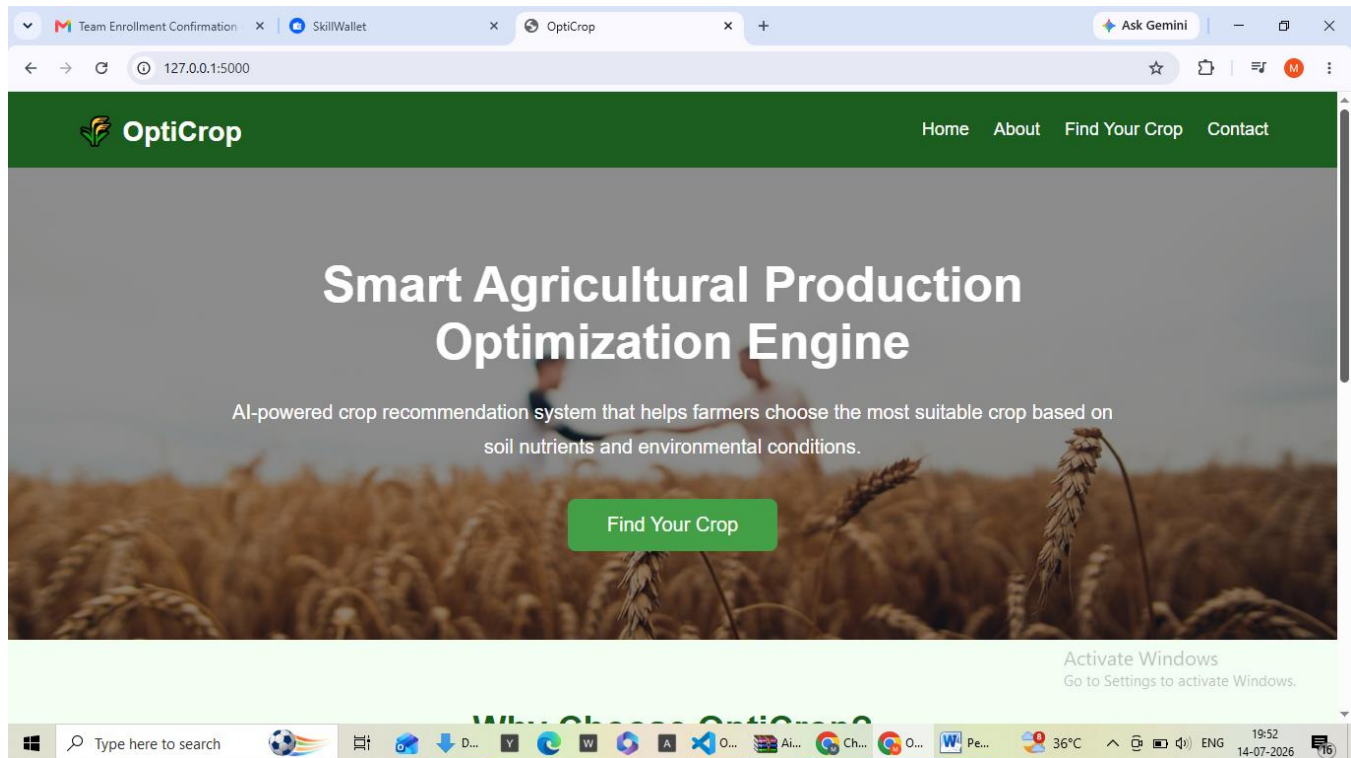
Description:

The Home Page serves as the entry point of the OptiCrop web application. It provides users with an attractive and user-friendly interface that introduces the purpose of the project. The homepage highlights the importance of Artificial Intelligence in modern agriculture and explains how OptiCrop helps farmers make informed crop selection decisions based on environmental and soil parameters.

The navigation bar allows users to easily access different sections of the application, including Home, About, Find Your Crop, and Contact. A prominent **"Find Your Crop"** button encourages users to navigate directly to the prediction module. The responsive design ensures that the application can be accessed conveniently on different screen sizes.

Home Page Features

- Responsive and user-friendly interface.
- Project introduction and overview.
- Navigation menu for easy access.
- Direct access to the crop prediction page.
- Modern and attractive design.



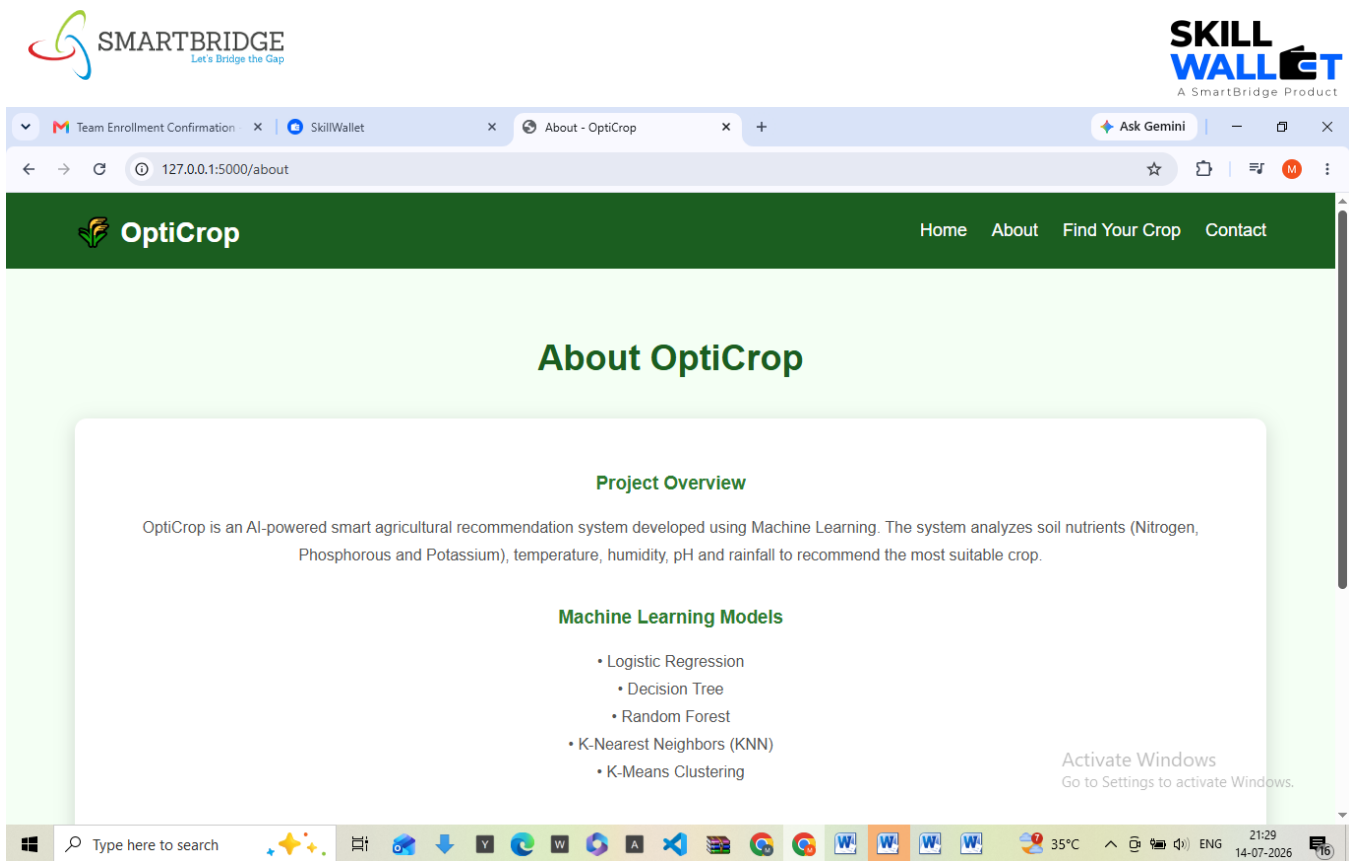
About Page

The About Page provides detailed information about the OptiCrop project and its objectives. It explains the motivation behind developing the application and describes how Machine Learning techniques are used to recommend suitable crops based on soil nutrients and environmental conditions.

The page also introduces the technologies used in the project, including Python, Flask, HTML, CSS, JavaScript, Scikit-learn, Pandas, and NumPy. It emphasizes the project's goal of supporting smart agriculture by improving crop productivity and helping farmers make data-driven cultivation decisions.

About Page Features

- Project overview.
- Objectives of OptiCrop.
- Information about Machine Learning integration.
- Technologies used.
- Benefits of AI-powered crop recommendation



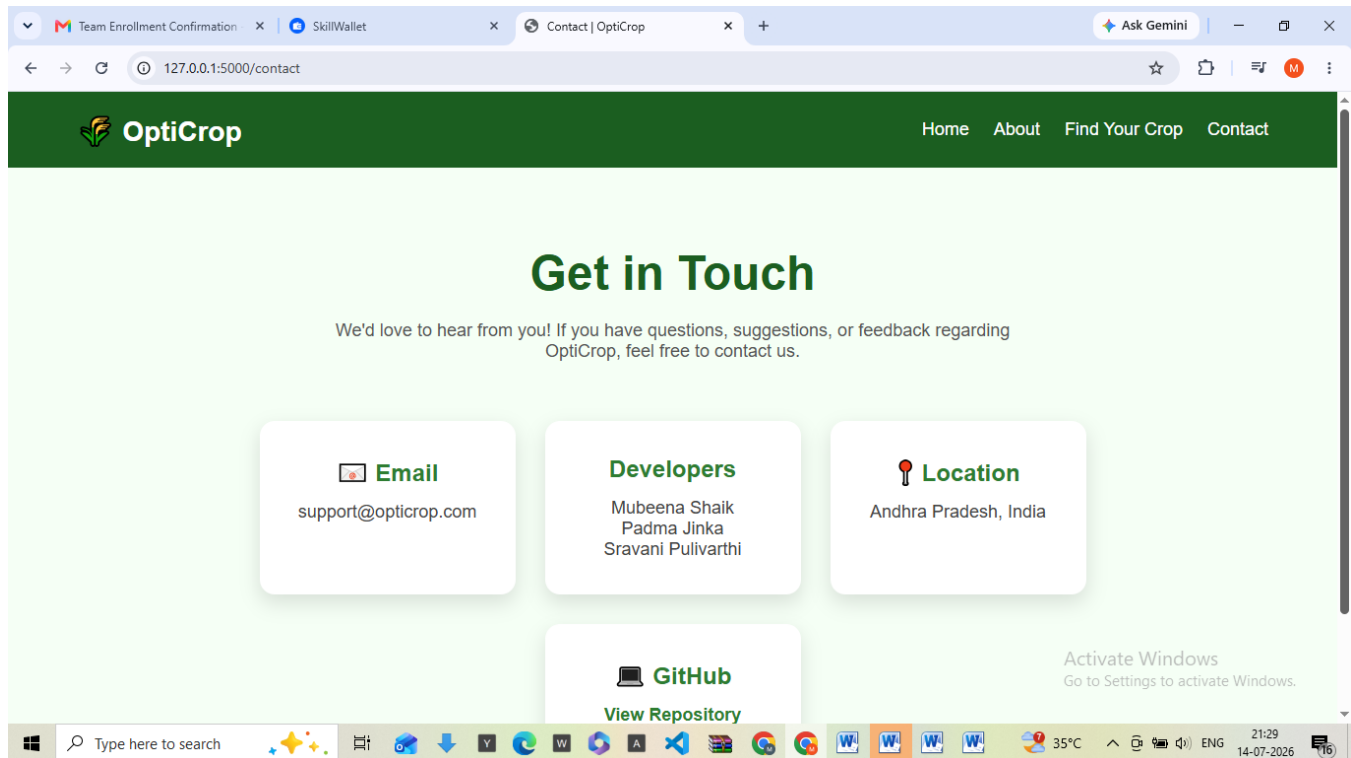
Contact Page

The Contact Page enables users to communicate with the development team by submitting their queries, suggestions, or feedback regarding the application. It provides a simple and intuitive contact form where users can enter their name, email address, subject, and message.

This feature encourages user interaction and allows valuable feedback to be collected for future improvements. The page is designed with a clean layout to provide a smooth user experience.

Contact Page Features

- Contact form for user queries.
- Name, Email, Subject, and Message fields.
- Easy-to-use interface.
- Supports user feedback and suggestions.
- Responsive design.



Find Your Crop Page

Crop Prediction Page

The **Find Your Crop** page is the core functionality of the OptiCrop application. Users enter agricultural parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall into the prediction form.

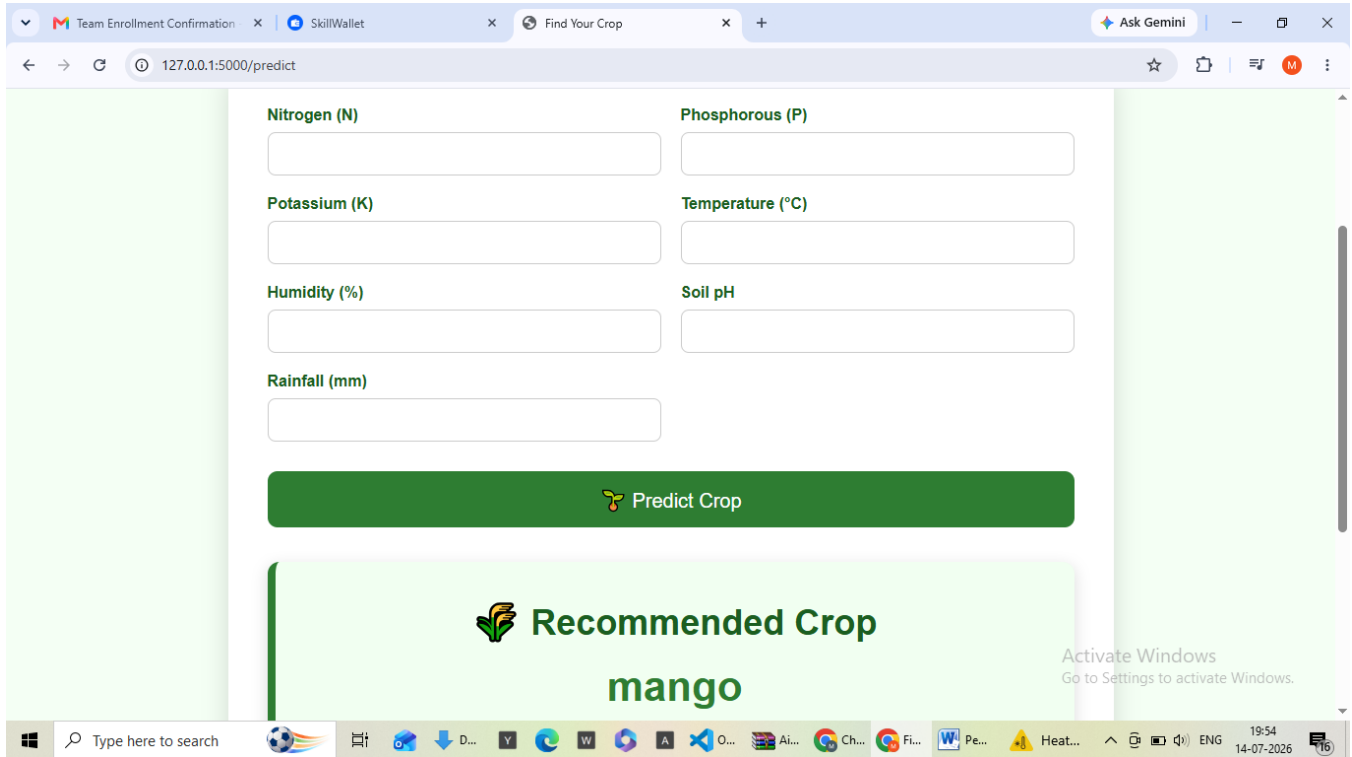
After submitting the values, the Flask backend processes the request and passes the data to the trained Random Forest Machine Learning model. The model predicts the most suitable crop and displays detailed cultivation information including the recommended season, soil type, water requirement, and temperature range.

Prediction Page Features

- Input form for seven agricultural parameters.
- Real-time crop prediction.
- Random Forest Machine Learning model integration.
- Displays crop details and cultivation information.
- Fast and accurate recommendations.

Find Your Crop Page

The Find Your Crop page allows users to enter soil nutrients and environmental parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall. These values are submitted to the prediction model.



The screenshot shows a web browser window with the URL `127.0.0.1:5000/predict`. The page has a light green background. It contains several input fields for user data:

- Nitrogen (N)**:
- Phosphorous (P)**:
- Potassium (K)**:
- Temperature (°C)**:
- Humidity (%)**:
- Soil pH**:
- Rainfall (mm)**:

Below the input fields is a large green button with a leaf icon and the text **Predict Crop**.

Below the button is a light green box with a leaf icon and the text **Recommended Crop** followed by **mango** in a larger font.

At the bottom right of the page, there is a small text overlay: **Activate Windows** Go to Settings to activate Windows.

The browser's taskbar at the bottom shows various application icons and the system clock indicating 19:54 on 14-07-2026.

Prediction Result

After entering the required values, the machine learning model predicts the most suitable crop and displays the recommendation instantly.

Team Enrollment Confirmation x SkillWallet x Find Your Crop x + Ask Gemini

127.0.0.1:5000/predict

Nitrogen (N)

Potassium (K)

Humidity (%)

Rainfall (mm)

Phosphorous (P)

Temperature (°C)

Soil pH

Predict Crop

 **Recommended Crop**
mango

Activate Windows
Go to Settings to activate Windows.

Type here to search

D... Y E W A O... AI... Ch... Fi... Pe... Heat... ENG 19:54 14-07-2026